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# An Intelligent Agent for Phishing Simulation and Behavioral Threat Analysis

*SemanticShield AI: A Behavior-Driven Framework for Realistic Phishing Simulation*

| 4 | 3 | 4 | 5 |
|---|---|---|---|
| BEHAVIORAL EMPLOYEE PROFILES | WEIGHTED ANALYSIS AGENTS | PHISHING ATTACK SCENARIOS | ARCHITECTURE LAYERS |

**PROJECT TEAM**

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## ABSTRACT

Phishing remains the most common entry point for breaches, yet most awareness programs rely on static, one-size-fits-all test emails that ignore how differently people actually react. This work presents SemanticShield AI, an intelligent agent that simulates phishing attacks against AI-driven employee personas, each with its own behavioral profile, and analyzes the resulting decisions. A Gemini-powered agent reasons through a phishing email as a trusting, security-conscious, overwhelmed or executive employee, streams its reasoning live, and commits to a verdict. The verdict is passed through a weighted multi-agent risk engine to produce a threat category, severity and confidence score, which are logged as incidents on a dashboard.

## RESEARCH GAPS ADDRESSED

- Static phishing tests that ignore individual behavior
- Simulation and threat analysis handled as separate tools
- No visible reasoning behind why a user falls for or spots an attack
- Generic risk scores that are not tied to employee role or profile
- Costly commercial platforms that are hard to adapt for research

## PHISHING SIMULATION PIPELINE: METHODOLOGY

**Scenario Input**
Scenario, employee role, behavior profile, email subject and body

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**Simulation Engine**
Central orchestrator · Google Gemini API

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I **Trusting Persona** (HR Manager)
II **Security-Conscious** (Developer)
III **Overwhelmed** (Finance Manager)
IV **Executive Persona** (VP)
V **Decision Extraction** [VULNERABLE / SAFE]
VI **Risk Scoring** Sem 40% · Beh 25% · URL 35%

→

**Threat Analysis Result**
Category, severity and confidence; logged as an incident and streamed live

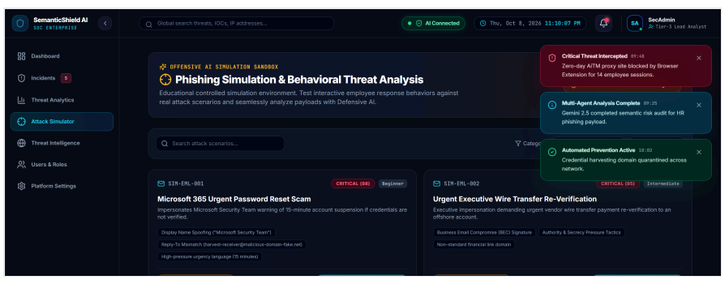
## LAYERED SYSTEM ARCHITECTURE

| | |
|---|---|
| **Presentation** | React two-panel interface · live reasoning stream |
| **Application** | FastAPI REST and WebSocket endpoints · Pydantic validation |
| **Business Logic** | Simulation engine, supervisor agent, risk engine, incident and activity managers |
| **AI Integration** | Google Gemini API · behavioral system prompts · streaming |
| **Data Persistence** | SQLite · SQLAlchemy · Alembic migrations |

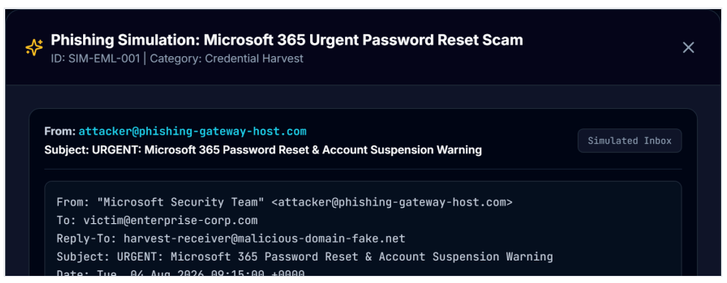
## END-TO-END WORKFLOW

1. Select Scenario
2. Choose Role & Profile
3. Load Phishing Email
4. Run Simulation
5. Gemini Streams Reasoning
6. Extract Decision
7. Score Risk
8. Log Incident & Activity
9. View Live Results

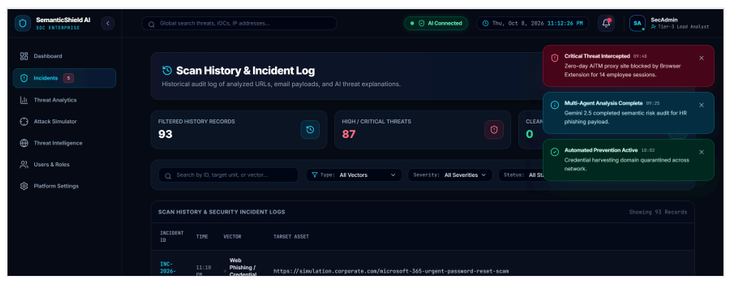
## IMPLEMENTATION: WORKING PROTOTYPE




Simulation Setup: scenario, role and profile selection




Live Analysis: streaming agent reasoning




Result Card: decision, category, severity, confidence

## PROFILES & SCENARIOS EVALUATED 4 profiles × 4 scenarios

- ✓ Trusting HR Manager
- ✓ Security-Conscious Developer
- ✓ Overwhelmed Finance Manager
- ✓ Executive VP
- ✓ Microsoft 365 Credential Harvest
- ✓ Wire Transfer Fraud
- ✓ Executive Impersonation
- ✓ QR Code Phishing

## FUNCTIONAL TESTS PASSED

- ✓ Simulation API [CONFIRM]
- ✓ Streaming Response [CONFIRM]
- ✓ Decision Extraction [CONFIRM]
- ✓ Risk Scoring [CONFIRM]
- ✓ Incident Creation [CONFIRM]
- ✓ Activity Logging [CONFIRM]
- ✓ WebSocket Updates [CONFIRM]
- ✓ Database Persistence [CONFIRM]

## TECHNOLOGY STACK

React.js · FastAPI · Python 3.10 · Gemini API · SQLite · SQLAlchemy · Alembic · WebSocket · Pydantic

## CONCLUSION

SemanticShield AI shows that the same phishing email carries very different risk depending on who receives it. By pairing behavior-driven simulation with a weighted multi-agent risk engine, it gives security teams a transparent, explainable view of human vulnerability and a practical basis for targeted awareness training.

## LIMITATIONS & FUTURE WORK

**Limits:** persona behavior depends on prompt design and the underlying language model; simulated personas are not real human responses; free-tier API rate limits; no large-scale user validation yet.

**Next:** calibrate personas with real employee response data, add SMS and voice phishing scenarios, adaptive training recommendations, attacker-versus-defender multi-agent loops, and migration to a cloud database.